LPO 8852: Regression II
Vanderbilt University
At-Home Final Exam
December 5, 2025

Name: _____

By signing below, I agree to the terms of Vanderbilt University's honor code. I attest that I have not collaborated with, or received any external assistance from other individuals or artificial intelligence tools on this at-home exam.

Signature: _____

Instructions: Read each question carefully and provide clear, concise responses in your own document. Be sure to complete every part of every question. Partial credit will be given where appropriate. If you make any assumptions to answer a question, please state those assumptions explicitly. Email your completed exam to sean.corcoran@vanderbilt.edu before 11:59 p.m. on **Saturday December 6**. Good luck!

Question 1. After Hurricane Katrina, the State of Louisiana took over the New Orleans public school district and enacted a wide set of school reforms, including converting most schools to privately-operated charter schools. One question is how these reforms affected overall school spending in New Orleans and spending on certain inputs, such as instructional and administrator salaries. Note these outcomes (e.g., total, instructional, and administrative spending per pupil) are all reported at the district level. (**23 points**)

(a) One approach to this research question might be a difference-in-differences design. Briefly describe the data requirements, estimating equation, and assumptions necessary for this approach. (**5 points**)

(b) An alternative approach would be a synthetic control design. When and why might this be preferable to the study you described in part (a)? (**3 points**)

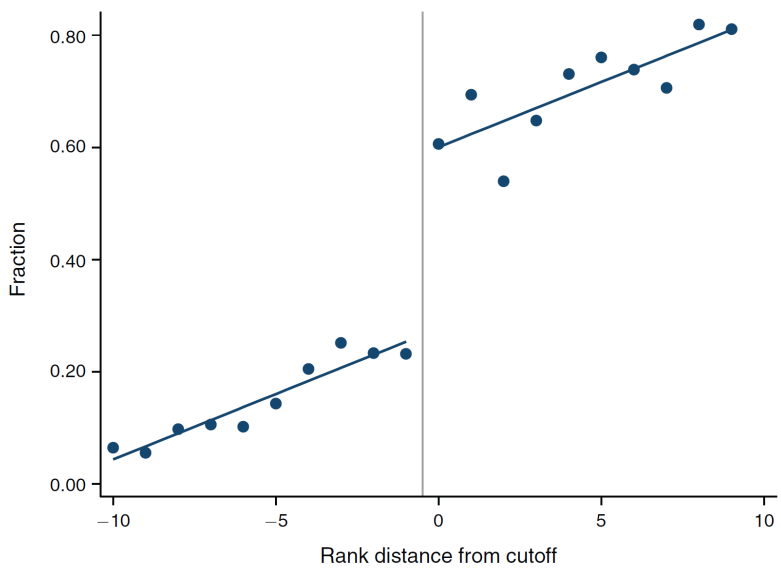
(c) Briefly describe how you would carry out a synthetic control study of post-Katrina public education spending, and any assumption(s) that must hold in order to interpret its findings as causal. (**5 points**)

(d) Suppose that your approach in (c) found that spending in New Orleans *increased* in post-Katrina years, relative to other districts. How would you determine whether this difference was statistically significant or not? Briefly explain. (**5 points**)

(e) State whether the following statement is <u>true</u> or <u>false</u>, and briefly justify your answer. (**5 points**)

When conducting a synthetic control analysis with multiple outcomes (e.g., total expenditure per pupil and administrative salaries per pupil), it is important to use the same synthetic control group for each outcome.

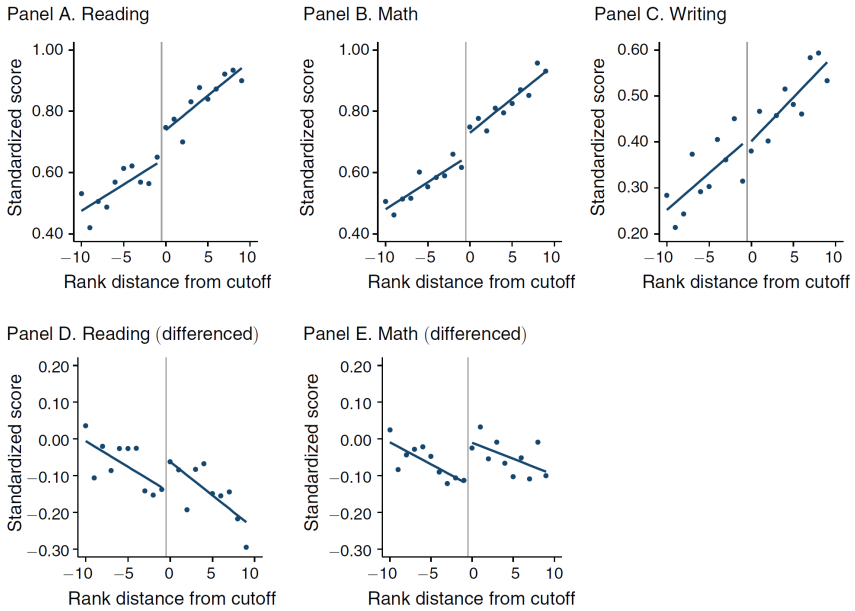
Question 2. A well-known paper by Card & Giuliano (2016) examined the effects of a tracking program on high-achieving Black and Hispanic students in a large urban school district. In the district they studied, schools with at least one gifted 4th grader were required to create a separate "gifted/high achiever" (GHA) classroom. Most seats in these classrooms were filled by non-gifted students who ranked highest on the previous year's end-of-grade test. The following figure shows the relationship between 3rd grade class rank and 4th grade assignment to a GHA classroom: (**36 points**)



(a) Briefly explain how these data might be used to support a regression discontinuity design analysis of the effects of GHA classroom assignment on later outcomes. (**5 points**)

(b) What are the key assumptions required for causal interpretation of the modeling approach described in part (a). What might you do, if anything, to evaluate the plausibility of those assumptions? Carefully explain. (**8 points**)

(c) The figure below was created using 4th grade test scores (shown on the y-axis). Consider the first two of these, for reading and math scores. True or false: the jump observed at $rank = 0$ is an estimate of the average treatment effect (ATE) of GHA classroom assignment. Explain your answer. (**5 points**)



(d) Carefully interpret the coefficient estimates in row 1, columns (1)-(5) below from the paper's Table 2 below. What evidence do columns 1 and 2 provide regarding the identifying assumptions for causal inference? (**8 points**)

TABLE 2—REGRESSION DISCONTINUITY ESTIMATES FOR FOURTH-GRADE OUTCOMES

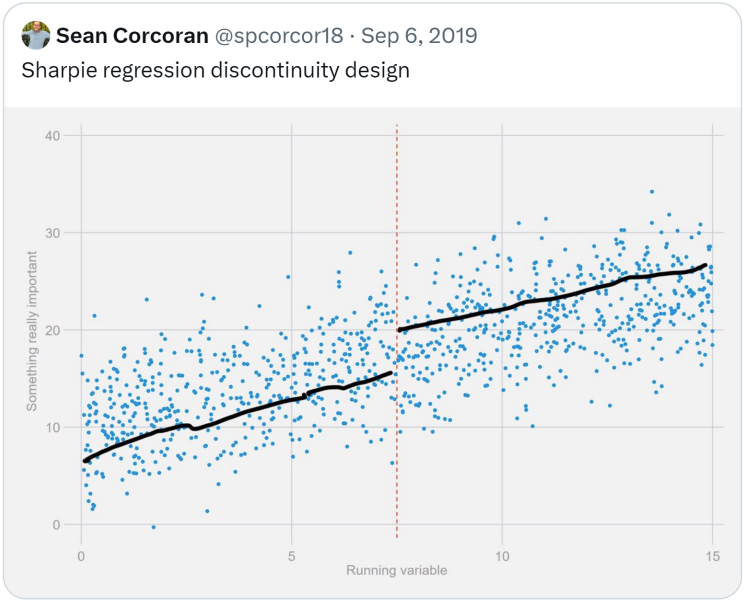
	Baseline achievement		First stage	Reduced-form outcomes		
	Third-grade reading (1)	Third-grade math (2)	Probability in GHA classroom (3)	Fourth-grade reading (4)	Fourth-grade math (5)	Fourth-grade writing (6)
1. No controls	0.008	−0.046	0.323	0.092	0.073	−0.011
	(0.029)	(0.043)	(0.025)	(0.034)	(0.039)	(0.054)
2. School and year fixed effects and student controls	0.015	−0.044	0.319	0.093	0.087	−0.012
	(0.027)	(0.040)	(0.026)	(0.031)	(0.035)	(0.051)
3. Differenced specification (based on change in test scores)	—	—	—	0.092	0.105	—
				(0.033)	(0.041)	
Observations	4,144	4,144	4,144	4,144	4,144	4,144

Notes: Estimates from models of dependent variable in column heading as a function of a student's rank (within school/cohort) on third-grade test scores. Entries are estimated coefficients on a dummy for the student's rank exceeding the cohort-specific cutoff for placement in the fourth-grade GHA classroom. All models include a linear term in rank interacted with the dummy. Models in rows 2 and 3 control for student demographics (age, gender, race/ethnicity, and median household income in zip code), dummies for year in fourth grade, and a complete set of school dummies. Models in row 2, columns 3–6 also control for third-grade scores in math and reading. Analysis sample is described in column 2 of Table 1. Parentheses contain standard errors, clustered by school.

(e) The coefficients in row 2 columns 4-5 (school and year fixed effects and student controls) are very similar to those in row 1. What might explain this, and what evidence does this provide for the identifying assumptions for causal inference? (**5 points**)

(f) (Unrelated to parts a-e): State whether the following statement is true or false. If false, explain why. (**5 points**)

If there is evidence of manipulation in the running variable in a regression discontinuity design, the treatment effect estimator will be biased.



Question 3. Proponents of school choice argue that public schools will improve with greater competition from charter and private schools. The theory is that, in the absence of competition, traditional public schools have weak incentives to perform at their highest levels. Competition raises the risk of losing students (and funding), and public schools respond by increasing effort, using resources more efficiently, or prioritizing different outcomes. Empirically estimating the causal effects of competition on student outcomes and school performance is challenging, however. (**15 points**)

(a) Consider a cross-sectional regression model of the following type:

$$y_{is} = \beta_0 + \beta_1 CharterComp_s + \gamma' X_i + \delta' W_s + u_{is}$$

in which y_{is} is an outcome for student i in traditional public school s, X_i is a vector of student-level covariates, and W_s is a vector of school-level characteristics. $CharterComp_s$ is a measure of charter school competition in the vicinity of school s; for example, this measure could be the number of charter schools within 1 mile of school s. The student outcomes (y_{is}) may include, for example, test scores or attendance. The coefficient of interest is β_1, the relationship between student outcomes at traditional schools and the extent of local charter school competition. Assume all schools in the data are located within the same large urban district.

What key assumption(s) must hold in order to interpret β_1 as the causal effect of charter competition in this model? If there is suspected bias, in what direction do you believe it goes? Carefully explain. (**5 points**)

(b) In this school district, charter schools are responsible for finding their own space, and are more likely to locate in neighborhoods where space can be found. These spaces often include former schools, churches, strip malls, or industrial space. Consider the following instrumental variable for $CharterComp_{st}$: the number of nearby buildings that are large enough to house a charter school. For example, if $CharterComp_{st}$ is the number of charter schools within a 1 mile radius of school s, the instrument z_{st} is the number of large buildings suitable for housing a charter school (say 30,000-60,000 square feet) within a 1 mile radius of school s. The estimating equation is otherwise the same as part (a), with the exception of the instrumental variable.

What are the assumptions necessary for z_{st} to be a valid instrument for $CharterComp_{st}$ in this application? Do you think these assumptions are likely to hold in this context? Carefully explain why or why not. (**5 points**)

(c) State whether the following statement is <u>true</u> or <u>false</u>. Briefly justify your answer. (**5 points**)

If a researcher includes an instrumental variable z in a causal regression model of interest and obtains a statistically significant coefficient for z, the instrument is invalid.